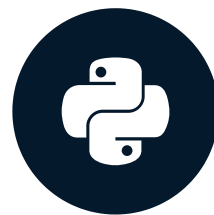


Welcome to Python!

PYTHON FOR SPREADSHEET USERS

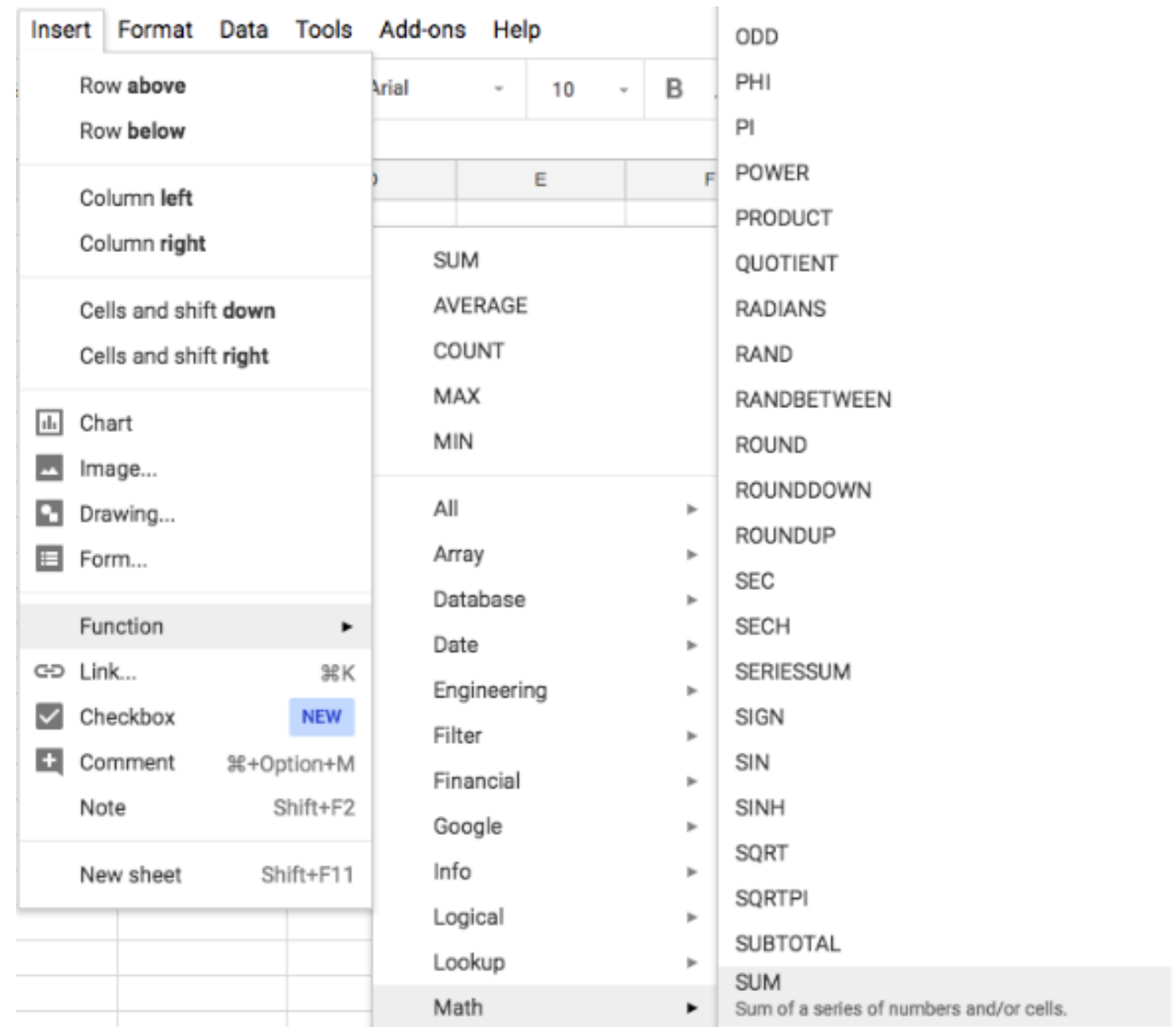


Chris Cardillo
Data Scientist

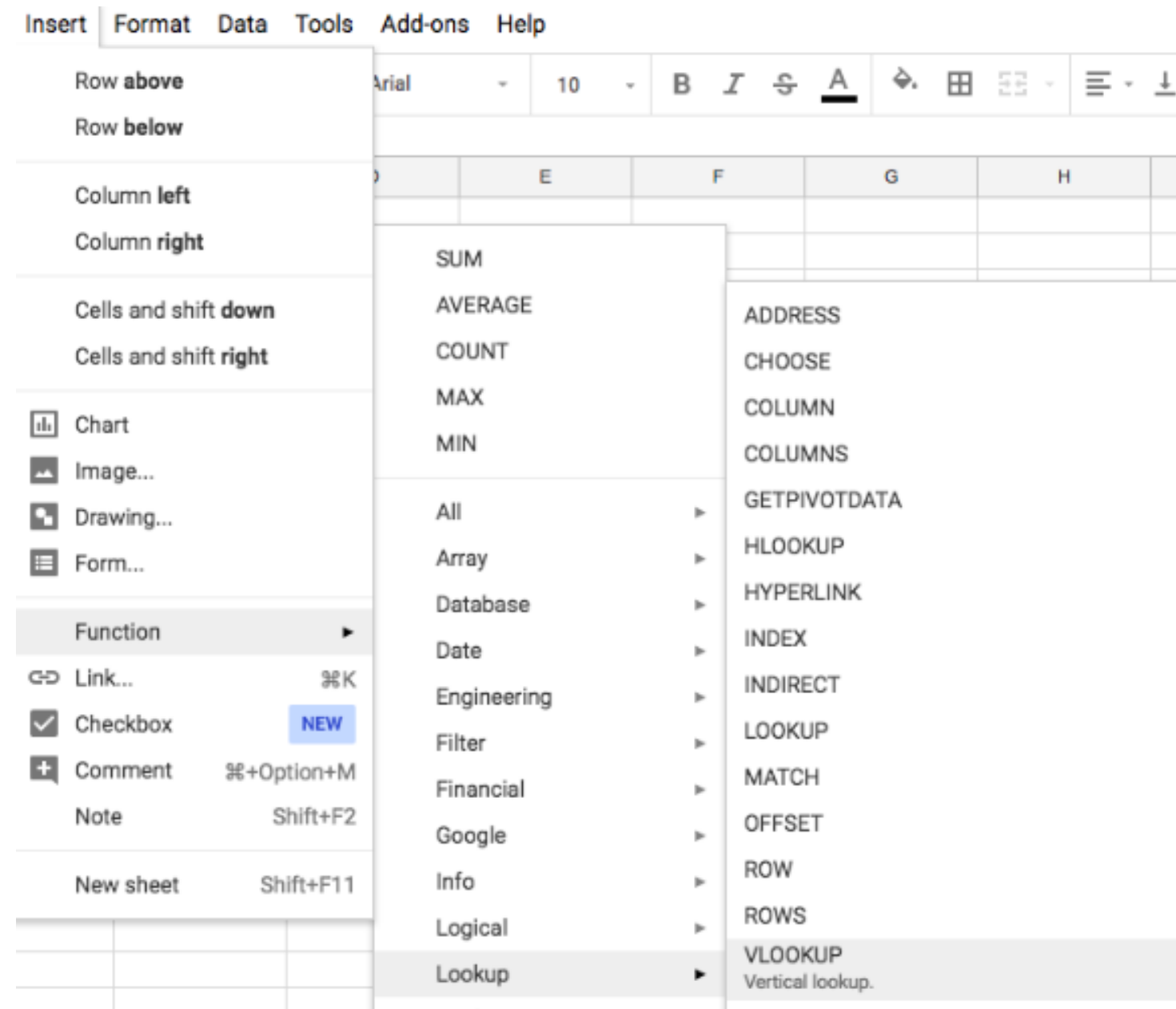
What is Python?

- General purpose programming language
- Has code packages for data science
- **Object-oriented**

Object-oriented spreadsheets



Object-oriented spreadsheets



Object-oriented Python

In Python

```
math.SUM(2, 3)
```

In Spreadsheets

 **Example Sheet**  

File Edit View Insert Format

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=SUM(2, 3)

	A	B	C
1	5		

The power of the dot

```
math.SUM()
```

- `math` is our package
- `SUM` is our function in the package
- The dot (`.`) lets us access the function inside the package

Scripts

A Recipe for Buttered Toast

1. Turn on toaster
2. Put bread in toaster for 1 minute
3. Take bread out of toaster
4. Put butter on toast

Importing packages

```
import math
```


Importing pandas



Importing pandas

```
import pandas
```

Loading Excel data

```
import pandas
```

```
pandas.read_excel('fruit.xlsx')
```

	A	B	C
1	name	color	price_usd
2	Apple	red	0.88
3	Banana	yellow	0.23
4	Orange	orange	0.68
5	Watermelon	green	3.98
6	Plum	purple	0.96

- **fruit.xlsx**
 - (the kind of data pandas likes)

pandas as pd

before

```
import pandas
```

```
pandas.read_excel('fruit.xlsx')
```

after

```
import pandas as pd
```

```
pd.read_excel('fruit.xlsx')
```

Variable assignment

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')
```

	A	B	C
1	name	color	price_usd
2	Apple	red	0.88
3	Banana	yellow	0.23
4	Orange	orange	0.68
5	Watermelon	green	3.98
6	Plum	purple	0.96

fruit.xlsx

Looking at data with print()

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')

print(fruit)
```

	name	color	price_usd
0	Apple	red	0.88
1	Banana	yellow	0.23
2	Orange	orange	0.68
3	Watermelon	green	3.98
4	Plum	purple	0.96
5	Blueberries	blue	5.16
6	Dragonfruit	pink	5.27
7	Kiwi	brown	1.12

Script and console panes in DataCamp

```
script.py
1  # Import pandas library
2  import pandas as pd
3
4  # Import the data
5  fruit = pd.read_excel('fruit.xlsx')
6
7  # Print the data
8  print(fruit)
```

```
IPython Shell  Slides
name  color  price_usd
0    Apple   red      0.88
1   Banana yellow    0.23
2    Orange orange    0.68
3 Watermelon green    3.98
4      Plum  purple    0.96
5 Blueberries  blue    5.16
6 Dragonfruit  pink    5.27
7      Kiwi   brown    1.12
```

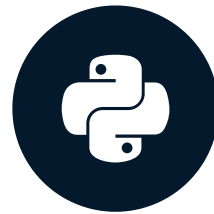
- `script.py` is the script
- `IPython Shell` is the console

Your turn!

PYTHON FOR SPREADSHEET USERS

DataFrames and their methods

PYTHON FOR SPREADSHEET USERS



Chris Cardillo
Data Scientist

Where we left off

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')

print(fruit)
```

	name	color	price_usd
0	Apple	red	0.88
1	Banana	yellow	0.23
2	Orange	orange	0.68
3	Watermelon	green	3.98
4	Plum	purple	0.96
5	Blueberries	blue	5.16
6	Dragonfruit	pink	5.27
7	Kiwi	brown	1.12

Anatomy of a pandas DataFrame

	name	color	price_usd
0	Apple	red	0.88
1	Banana	yellow	0.23
2	Orange	orange	0.68
3	Watermelon	green	3.98
4	Plum	purple	0.96
5	Blueberries	blue	5.16
6	Dragonfruit	pink	5.27
7	Kiwi	brown	1.12

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5	Blueberries	blue	5.16
6	Dragonfruit	pink	5.27
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3	Watermelon	green	3.98
4	Plum	purple	0.96
5	Blueberries	blue	5.16
6	Dragonfruit	pink	5.27
7	Kiwi	brown	1.12

DataFrame methods

- `.head()`
- `.info()`
- `.describe()`
- `.sort_values()`

The .head() method

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')

print(fruit.head())
```

	name	color	price_usd
0	Apple	red	0.88
1	Banana	yellow	0.23
2	Orange	orange	0.68
3	Watermelon	green	3.98
4	Plum	purple	0.96

The .head() method

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')

print(fruit.head(2))
```

	name	color	price_usd
0	Apple	red	0.88
1	Banana	yellow	0.23

The .info() method

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')

print(fruit.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
--  --
0   name        8 non-null     object
1   color       8 non-null     object
2   price_usd   8 non-null     float64
dtypes: float64(1), object(2)
memory usage: 272.0+ bytes
```

The .describe() method

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')

print(fruit.describe())
```

	price_usd
count	8.000000
mean	2.285000
std	2.135911
min	0.230000
25%	0.830000
50%	1.040000
75%	4.275000
max	5.270000

The .sort_values() method

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')

fruit = fruit.sort_values('name')
fruit = fruit.reset_index(drop=True)

print(fruit)
```

	name	color	price_usd
0	Apple	red	0.88
1	Banana	yellow	0.23
2	Blueberries	blue	5.16
3	Dragonfruit	pink	5.27
4	Kiwi	brown	1.12
5	Orange	orange	0.68
6	Plum	purple	0.96
7	Watermelon	green	3.98

The .sort_values() method

```
import pandas as pd

fruit = pd.read_excel('fruit.xlsx')

fruit = fruit.sort_values('price_usd', ascending=False)
fruit = fruit.reset_index(drop=True)

print(fruit.head(3))
```

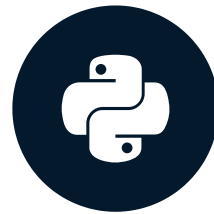
	name	color	price_usd
0	Dragonfruit	pink	5.27
1	Blueberries	blue	5.16
2	Watermelon	green	3.98

Your turn!

PYTHON FOR SPREADSHEET USERS

Filtering rows and creating columns

PYTHON FOR SPREADSHEET USERS



Chris Cardillo
Data Scientist

Filtering in spreadsheets

	A	B	C	D	
1	name ▼	color ▾	price_usd ▾		
2	Apple				
10		Sort A → Z			
11		Sort Z → A			
12					
13		▸ Filter by condition...			
14		▼ Filter by values...			
15		Select all - Clear			▼
16					
17					🔍
18		✓ Apple			
19		Banana			
20		Blueberries			
21		Dragonfruit			
22		Kiwi			
23					
24					
25					
26					
27					

Accessing a single column

	name	color	price_usd
0	Apple	red	0.88
1	Banana	yellow	0.23
2	Orange	orange	0.68
3	Watermelon	green	3.98
4	Plum	purple	0.96
5	Blueberries	blue	5.16
6	Dragonfruit	pink	5.27
7	Kiwi	brown	1.12

Accessing a single column

```
fruit['name']
```

```
0      Apple
1     Banana
2     Orange
3  Watermelon
4       Plum
5  Blueberries
6  Dragonfruit
7        Kiwi
Name: name, dtype: object
```

Comparison operators

- `==` (equal to)
- `!=` (not equal to)
- `<` (less than)
- `>` (greater than)
- `<=` (less than or equal to)
- `>=` (greater or equal to)

Comparisons

```
fruit['name']
```

```
0      Apple
1     Banana
2     Orange
3  Watermelon
4        Plum
5  Blueberries
6  Dragonfruit
7        Kiwi
Name: name, dtype: object
```

```
fruit['name'] == "Apple"
```

```
0      True
1     False
2     False
3     False
4     False
5     False
6     False
7     False
Name: name, dtype: bool
```


Filtering

```
fruit[fruit['name'] == "Apple"]
```

	name	color	price_usd
0	Apple	red	0.88

Filtering

```
fruit[fruit['price_usd'] > 1]
```

	name	color	price_usd
3	Watermelon	green	3.98
5	Blueberries	blue	5.16
6	Dragonfruit	pink	5.27
7	Kiwi	brown	1.12

Filtering

```
fruit[fruit['price_usd'] > 1].reset_index(drop=True)
```

	name	color	price_usd
0	Watermelon	green	3.98
1	Blueberries	blue	5.16
2	Dragonfruit	pink	5.27
3	Kiwi	brown	1.12

Basic filtering pattern

```
df[df['this_column'] == 'that value']
```

Basic filtering pattern

In Python

```
fruit[fruit['name'] == "Apple"]
```

```
      name  color  price_usd
0  Apple    red      0.88
```

In Spreadsheets

	A	B	C	D	
1	name ▼	color ≡	price_usd ≡		
2	Apple				
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					
26					
27					

Sort A → Z

Sort Z → A

► Filter by condition...

▼ Filter by values...

[Select all](#) - [Clear](#)

✓ Apple

Banana

Blueberries

Dragonfruit

Kiwi

OK Cancel

Creating a new column

	name	color	price_usd
0	Apple	red	0.88
1	Banana	yellow	0.23
2	Orange	orange	0.68
3	Watermelon	green	3.98
4	Plum	purple	0.96
5	Blueberries	blue	5.16
6	Dragonfruit	pink	5.27
7	Kiwi	brown	1.12

Creating a new column

				<i>fx</i> <code>=C5*2</code>
	A	B	C	D
1	name	color	price_usd	cost
2	Apple	red	0.88	1.76
3	Banana	yellow	0.23	0.46
4	Orange	orange	0.68	1.36
5	Watermelon	green	3.98	<code>=C5*2</code>
6	Plum	purple	0.96	1.92
7	Blueberries	blue	5.16	10.32
8	Dragonfruit	pink	5.27	10.54
9	Kiwi	brown	1.12	2.24

Mathematical operators

- `+` (addition)
- `-` (subtraction)
- `*` (multiplication)
- `/` (division)

Creating a new column

```
fruit['cost'] = fruit['price_usd'] * 2
```

	name	color	price_usd	cost
0	Apple	red	0.88	1.76
1	Banana	yellow	0.23	0.46
2	Orange	orange	0.68	1.36
3	Watermelon	green	3.98	7.96
4	Plum	purple	0.96	1.92
5	Blueberries	blue	5.16	10.32
6	Dragonfruit	pink	5.27	10.54
7	Kiwi	brown	1.12	2.24

Creating a new column

	name	color	price_usd	quantity
0	Apple	red	0.88	3
1	Banana	yellow	0.23	2
2	Orange	orange	0.68	2
3	Watermelon	green	3.98	1
4	Plum	purple	0.96	0
5	Blueberries	blue	5.16	1
6	Dragonfruit	pink	5.27	1
7	Kiwi	brown	1.12	1

Creating a new column

fx | `=D5*C5`

	A	B	C	D	E
1	name	color	price_usd	quantity	cost
2	Apple	red	0.88	3	2.64
3	Banana	yellow	0.23	2	0.46
4	Orange	orange	0.68	2	1.36
5	Watermelon	green	3.98	1	<code>=D5*C5</code>
6	Plum	purple	0.96	0	0
7	Blueberries	blue	5.16	1	5.16
8	Dragonfruit	pink	5.27	1	5.27
9	Kiwi	brown	1.12	1	1.12

3.98 ×

Creating a new column

```
fruit['cost'] = fruit['price_usd'] * fruit['quantity']
```

	name	color	price_usd	quantity	cost
0	Apple	red	0.88	3	2.64
1	Banana	yellow	0.23	2	0.46
2	Orange	orange	0.68	2	1.36
3	Watermelon	green	3.98	1	3.98
4	Plum	purple	0.96	0	0.00
5	Blueberries	blue	5.16	1	5.16
6	Dragonfruit	pink	5.27	1	5.27
7	Kiwi	brown	1.12	1	1.12

Your turn!

PYTHON FOR SPREADSHEET USERS